

Joshua Antu Kannappilly

jkannapp@nd.edu | (347) 776-2026 | Notre Dame, IN | Brooklyn, NY

EDUCATION

University of Notre Dame Bachelor of Science Major: Electrical Engineering Concentration: Semiconductors & Nanotechnology Relevant Coursework: Digital Integrated Circuits, Computer Architecture, Applied Embedded Systems, Analog and Digital Circuit Design, System Theory , Fundamentals of Semiconductors, Logic and Processor Design, Signals and Information Systems, Digital Design for Interconnected Systems, Electronic Devices and Systems	Notre Dame, IN May 2028 3.89/4.00
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EXPERIENCE

IrishSat – CubeSat Electrical Chief Engineer	Jan 2025 – Present
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- Successfully led the redesign of a custom Battery Management System (BMS) in KiCad using the Texas Instruments BQ40Z50-R2 to manage a 2S2P lithium-ion battery pack.
- Implemented SMBus communication with a central power management board for state-of-health monitoring and system integration.
- Diagnosed critical issues in the initial BMS iteration during board bring-up, including mismatched I2C logic levels (1.8V vs 3.3V), identifying the need for a new IC architecture.

NDnano – Undergraduate Research Fellow	May 2026 – July 2026
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- Co-authored 2 paper submissions to the IEEE International Electron Devices Meeting (IEDM) as 3rd author, investigating fluid/conventional imprint mechanisms, coercive voltage degradation, and charge-trapping kinetics in HZO-based FeRAM devices.
- Executed full fabrication workflows in a cleanroom environment, gaining hands-on operation of advanced processing equipment to successfully build high-*k* MOSFET devices from scratch.

Bernstein Lab, University of Notre Dame – Undergraduate Research Assistant	Sep 2025 – Dec 2025
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- Modernized a legacy Python library for the Newport SMC100 controller, implementing ESP-error handling and translating raw encoder counts into high-precision angular units (0.001° resolution).
- Integrated customized motion control drivers into the QCoDeS measurement framework, utilizing hardware abstraction to automate data logging into a centralized SQL database.
- Engineered a 2D “snaking” raster scan algorithm for Thorlabs Kinesis motors, optimizing spatial mapping efficiency and reducing mechanical wear for future LWIR thermal imaging.

Midi Synthesizer – Individual Project	Jul 2025 – Aug 2025
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- Engineered a MIDI Synthesizer, interfacing an ESP32 microcontroller with a USB Host Shield, PCM5102A DAC, and PAM8403D amplifier to drive dual 40mm speakers, enabling digital piano connectivity.
- Designed and routed a custom PCB in KiCad to integrate all components, optimizing for signal integrity and component layout for the compact synthesizer.
- Developed C++ firmware for the ESP32, implementing software-defined waveform generation for sine, square, triangle, and sawtooth waves, alongside real-time MIDI processing and polyphonic voice management.

LEADERSHIP

South Asian Student Association, University of Notre Dame – Vice President	August 2025 – Present
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- Drove a 20-member increase in signups for Jazba, the club’s signature performance showcase, by spearheading a suite of community-building initiatives that deepened member engagement year-round.
- Generated 103 signups at the Activities Fair and represented South Asian culture at the MSPS Fair through cultural games such as carrom expanding the club’s visibility across the broader Notre Dame community.

TECHNICAL SKILLS

Hardware: PCB Design, Embedded Systems (Arduino, ESP32, Raspberry Pi), Semiconductor Characterization
Programming: Python, C, C++, SQL, MATLAB, Ruby, JavaScript, RTOS, Serial (I²C, I²S, SPI), Automated Testing
Other: KiCad, Pandas, Matplotlib, Git, GitHub, Linux, SolidWorks, Rails, Automation Tools, Troubleshooting